Quantum-Assisted Multi-Asset Portfolio Construction

Step 05

Tunable Investor Goals

Mapping growth, income, drawdown control, and cost sensitivity into exact optimization

Project technical report

Contents

1 Why investor preferences need their own stage

A fixed objective hides an important fact: there is no universally correct balance among growth, income, downside control, and implementation cost. Step 5 exposes those priorities as four scores from 0 to 100 and maps them transparently into the same exact constrained optimizer.

2 Preference scores and normalized shares

Let the entered scores be

$$s = (s_g, s_y, s_d, s_c), \qquad 0 \le s_j \le 100. \tag{1}$$

They are converted into shares

$$\pi_j = \frac{s_j}{s_g + s_y + s_d + s_c}. \tag{2}$$

For the selected profile $(55, 55, 70, 60)$,

$$(\pi_g, \pi_y, \pi_d, \pi_c) = (0.229, 0.229, 0.292, 0.250). \tag{3}$$

The shares sum to one and describe relative preference, not expected return percentages.

3 Why scaling is necessary

Growth may be around 0.03, variance around 0.006, turnover around 0.2, and impact cost around 10^{-5}. Directly multiplying these quantities by the same preference score would make the largest numerical unit dominate.

Step 5 first constructs fully feasible anchors: current portfolio, Step 4 scenario-aware, maximum feasible growth, maximum feasible income, and minimum feasible variance. For component j, its scale is

$$S_j = \max\left\{\max(x_j) - \min(x_j),\ 0.1Q_{0.75}(|x_j|),\ \mathrm{floor}_j\right\}. \tag{4}$$

The normalized component x_j/S_j is then dimensionless and comparable across goals.

4 Splitting broad goals into model components

Drawdown control is represented by variance and scenario-warning hinge:

$$\pi_d\left(0.35\frac{\mathrm{Var}}{S_{\mathrm{Var}}} + 0.65\frac{\mathrm{Hinge}}{S_{\mathrm{Hinge}}}\right). \tag{5}$$

Cost sensitivity is split among linear cost, impact, and gross turnover:

$$\pi_c\left(0.30\frac{C_{\mathrm{lin}}}{S_{\mathrm{lin}}} + 0.25\frac{C_{\mathrm{imp}}}{S_{\mathrm{imp}}} + 0.45\frac{T}{S_T}\right). \tag{6}$$

This prevents “cost sensitivity” from meaning only commissions while ignoring implementation activity.

5 Derived objective coefficients

The coefficient mapping is

$$\alpha_g = \frac{\pi_g}{S_g}, \qquad \alpha_y = \frac{\pi_y}{S_y}, \tag{7}$$

$$\lambda_v = \frac{0.35\pi_d}{S_{\text{Var}}}, \qquad \lambda_s = \frac{0.65\pi_d + \epsilon_s}{S_{\text{Hinge}}}, \tag{8}$$

$$\lambda_L = \frac{0.30\pi_c + \epsilon_e}{S_{\text{lin}}}, \qquad \lambda_I = \frac{0.25\pi_c + \epsilon_e}{S_{\text{imp}}}, \tag{9}$$

$$\lambda_T = \frac{0.45\pi_c + \epsilon_e}{S_T}. \tag{10}$$

Small tie-breakers ϵ_s and ϵ_e preserve a preference for cleaner scenario and execution outcomes even when the corresponding user share is zero.

The effective Step 5 objective is

$$F = -\,\alpha_g G(w) - \alpha_y Y(w) + \lambda_v \text{Var}(w) + \lambda_s \text{Hinge}(w) \tag{11}$$

$$+\,\lambda_L C_{\text{lin}}(w) + \lambda_I C_{\text{imp}}(w) + \lambda_T T(w) + \lambda_c C(w). \tag{12}$$

All Step 4 hard constraints remain active.

6 Feasible anchors and calibration logic

The maximum-growth and maximum-income anchors are solved as linear programs with full budget, turnover, capacity, class, factor, floor, and hard-scenario constraints. Minimum feasible variance is solved as a fully constrained quadratic problem. Because every anchor is feasible, the scales reflect the actual investable opportunity set rather than arbitrary values.

7 Selected profile and strict-warning alternative

The primary selected profile uses the soft-warning policy: warning excess is penalized but not made a hard feasibility condition. A second strict-warning portfolio hardens the warning thresholds, requiring zero warning breaches. This creates a clean governance comparison.

Portfolio	Return	Volatility	Worst stress	Turnover	Effective holdings	Warnings
Current strategic	5.55%	8.82%	13.83%	0.00%	40.14	5
Primary classical	5.49%	7.86%	12.71%	13.09%	33.73	5
Strict warning	5.48%	7.51%	12.17%	23.88%	29.35	0

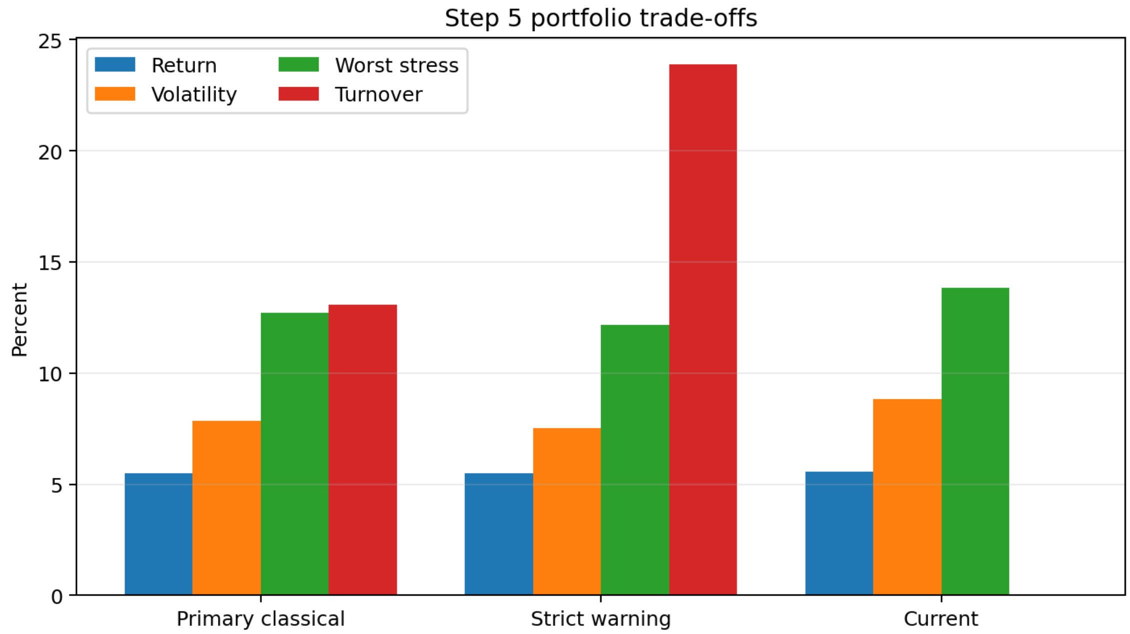

Figure 1: The strict-warning solution improves modeled downside risk at the cost of more turnover.

8 Interpreting the primary classical result

The primary portfolio has 5.49% expected return, 7.86% volatility, 12.71% worst modeled stress loss, 13.09% gross turnover, and 33.73 effective holdings. It is the best solution under the base Step 5 objective and later becomes the reference continuous optimum.

9 Interpreting the strict-warning result

The strict portfolio has nearly the same expected return, 5.48%, but lower volatility, lower worst stress loss, lower modeled drawdown, and zero soft-warning breaches. It requires 23.88% turnover and a slightly larger one-time cost. This is a financially sensible exchange: stronger governance is purchased with more implementation activity.

10 Directional sensitivity logic

A valid preference interface must move the solution in the intended direction. The one-way sensitivity procedure varies one score through 0, 25, 50, 75, and 100 while holding the others at 50. The checks require higher growth preference to increase growth exposure, higher income preference to increase income, higher drawdown preference to reduce risk measures, and higher cost sensitivity to reduce implementation burden. These are directional tests, not claims that every metric changes monotonically at every small increment.

11 Path-based diagnostics

For daily portfolio returns $r_{p,t}$, the wealth path is

$$W_t = \prod_{u \le t} (1 + r_{p,u}). \tag{13}$$

Maximum drawdown is

$$\mathrm{MDD} = \max_t \left(1 - \frac{W_t}{\max_{u \le t} W_u}\right). \tag{14}$$

Daily 95% VaR is the negative fifth percentile, and CVaR is the average loss in the tail below that percentile. In the final project these are synthetic-path diagnostics, not historical backtest statistics.

12 Reproduction checklist

Recompute feasible anchors, derive scales, transform scores into shares, calculate objective coefficients, solve with every Step 4 hard constraint active, audit exact hinge values, and save both primary and strict-warning solutions. Do not compare raw preference scores directly with returns or risk values.

13 What Step 5 establishes

Step 5 establishes a transparent preference-to-portfolio mechanism and produces the classical benchmarks used by the hybrid and validation stages. It does not establish that the strict portfolio is universally best; its advantage depends on the mandate and the challenge scoring definition.

A Why preference scores need calibration

An investor can say "growth 55, income 55, drawdown 70, cost 60," but the raw score numbers do not yet belong in an objective. Growth is measured in annual return, variance in squared return, and cost in one-time portfolio percentage. Step 5 first calibrates a characteristic scale for every metric. If $m_j(w)$ is metric j and s_j is its scale, the normalized metric is

$$\widetilde{m}_j(w) = \frac{m_j(w)}{s_j}. \tag{15}$$

User scores $u_j \ge 0$ are converted to preference shares

$$\pi_j = \frac{u_j}{\sum_k u_k}. \tag{16}$$

For scores $(55, 55, 70, 60)$, the shares are approximately $(22.9\%, 22.9\%, 29.2\%, 25.0\%)$.

B Goal-aware objective

A simplified normalized objective is

$$F_\pi(w,p,n,h) = -\pi_g \frac{g^\top w}{s_g} - \pi_y \frac{y^\top w}{s_y} + \pi_d \left[a_v \frac{w^\top \Sigma w}{s_v} + a_s \frac{\sum_s \omega_s h_s^2}{s_s} \right] \tag{17}$$

$$+ \pi_c \left[a_L \frac{c^\top (p+n)}{s_L} + a_I \frac{\Delta w^\top \Lambda \Delta w}{s_I} \right] + \lambda_{\text{conc}} \frac{w^\top w}{s_{\text{conc}}}. \tag{18}$$

The internal coefficients a_v, a_s, a_L, a_I distribute a high-level goal across related mathematical terms. Hard constraints remain unchanged. Thus a preference change alters the trade-off among feasible portfolios, not the definition of feasibility.

C How calibration anchors are constructed

The project solves several interpretable anchor portfolios: current, minimum variance, mean–variance, maximum feasible growth, maximum feasible income, and scenario-aware. The range of each metric across anchors supplies a scale and a sanity check. If changing the growth score does not improve expected growth, either the scale is wrong, the objective is dominated by another term, or the feasible set prevents movement.

D Directional sensitivity tests

Step 5 treats preference behavior as a testable requirement:

- increasing growth preference should not reduce the growth metric materially;
- increasing income preference should increase income yield;
- increasing drawdown-control preference should reduce volatility, drawdown, or scenario risk;
- increasing cost preference should reduce trading cost or turnover.

These are directional tests, not promises of strict monotonicity at every score because constraints can make the frontier piecewise and flat.

E Primary and strict-warning policies

The primary portfolio minimizes the base preference objective while allowing soft-warning excess to be penalized. The strict-warning portfolio changes the policy so warning thresholds behave as hard caps or receive sufficiently strong enforcement to eliminate warning breaches. The two portfolios answer different governance questions.

The primary result has approximately 5.49% expected return, 7.86% volatility, 12.71% worst scenario loss, 13.09% gross turnover, and five soft warnings. The strict result has approximately 5.48% expected return, 7.51% volatility, 12.17% worst scenario loss, 23.88% turnover, and zero soft warnings. The return difference is small relative to the modeled risk reduction, but the stricter policy requires more trading.

F Why a strict-warning portfolio can trade more

The current portfolio begins near or above several warning thresholds. Moving every scenario below its warning level can require coordinated changes across asset classes. Because warning constraints act simultaneously, the optimizer may need to sell several correlated equity exposures and add bonds or cash. The additional turnover is therefore an economic price of stronger governance rather than a numerical defect.

G Interpreting realized metrics in a synthetic run

The notebook also reports metrics calculated from the synthetic daily return sample, such as annualized realized return, annualized volatility, drawdown, daily VaR, and daily CVaR. These are in-sample realizations under the simulated data. They help compare candidates under common draws, but they are not historical backtest evidence.

For daily returns $r_{p,t}$, cumulative wealth is

$$W_t = \prod_{u \le t} (1 + r_{p,u}). \tag{19}$$

Drawdown is

$$D_t = 1 - \frac{W_t}{\max_{u \le t} W_u}, \qquad \mathrm{MDD} = \max_t D_t. \tag{20}$$

VaR at 95% is the negative fifth percentile of daily return, and CVaR is the mean loss conditional on being beyond that quantile.

H Profile reproducibility

For each profile, save the score vector, preference shares, scales, risk-policy mode, cost scenario, solver status, objective components, final weights, trades, constraint audit, and metrics. A profile name alone is not reproducible because the meaning of “balanced” depends on these numerical settings.

I Common interpretation errors

- Effective holdings is a count, not a percentage.
- One-time trading cost should not be compared directly with annual return without declaring a holding horizon.
- A soft-warning breach is not a hard-constraint failure.
- Similar expected return does not imply similar downside risk.
- A preference score is not an expected-return forecast.

J Reproduction protocol

1. Load the Step 4 context and calibration anchors.

2. Compute and store metric scales.
3. Normalize the user score vector into shares.
4. Construct the goal-aware objective without changing hard constraints.
5. Solve the requested profile and the strict-warning reference.
6. Recompute metrics and run all directional sensitivity checks.
7. Export allocation, trades, profile metrics, preference shares, and the full constraint audit.

Preference-semantics clarification

These notes align interpretation with the final executed code path and saved evidence without changing the reported numerical results.

- The four 0-100 inputs determine relative preference shares. Scaling all four scores by the same positive constant leaves the normalized shares unchanged.

- The label "Drawdown" is a user-facing shorthand for a downside-risk/drawdown-control proxy. The optimization primarily changes variance and scenario-warning penalties; path-dependent maximum drawdown is measured afterward rather than optimized directly.

- The scale construction is a calibrated multi-objective normalization based on feasible anchors. It is a transparent research utility specification, not a universal investor utility function.